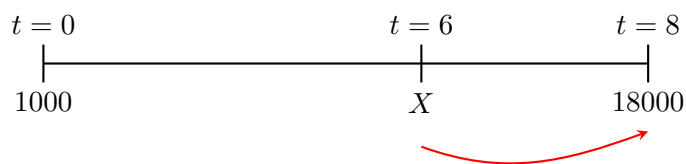


Problem 2.22

Suppose that $a(t) = 0.1t^2 + 1$. At time 0, \$1,000 is invested. An additional investment of \$X is made at time 6. If the total accumulated value of these two investments at time 8 is \$18,000, find X.

Solution

We organize the cash flows on the timeline:



First, compute the value of the initial investment at time 6:

$$A(6) = A(0) a(6).$$

Since $A(0) = 1000$ and

$$a(6) = 0.1(6)^2 + 1,$$

we get

$$A(6) = 1000[0.1 \cdot 36 + 1] = 1000(4.6) = 4600.$$

Also,

$$a(8) = 0.1(8)^2 + 1 = 7.4.$$

We will now find the value of x,

$$X \cdot \frac{a(8)}{a(6)} = 18000.$$

Let's not forget to subtract 4,600

$$X = \frac{18000 \cdot a(6)}{a(8)} - 4600.$$

Substituting the values,

$$X = \frac{18000 \cdot 4.6}{7.4} - 4600.$$

Therefore,

$$X = 6589.189189 \dots$$